

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and owning distributed systems at scale. Built search infrastructure for Google's core product, video serving systems at YouTube, and real-time marketplace platforms at Whatnot. Led teams of engineers across multiple organizations delivering high-traffic backend services. Expertise spans **distributed systems**, **search infrastructure**, **backend services**, **large-scale systems**, and **video platform** technologies. Managed cross-functional partnerships across product, infrastructure, and data teams. Seeking principal-level technical contributions on complex backend architecture challenges. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Distributed Systems & Backend: distributed systems, backend services, API design, microservices, large-scale systems, database systems

Programming Languages: Python, Go, C++, Java, SQL, type-safe APIs, TypeScript, JavaScript, HTML, CSS

Infrastructure: Kubernetes, container orchestration, caching systems, message queues, service mesh

Platforms: GCP, Linux, build systems, observability tooling

Technical Leadership: architecture design, technical direction, cross-functional collaboration, mentorship, Node.js, AWS

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designed video serving infrastructure handling billions of daily requests, optimizing caching strategies that reduced playback latency by 20%.
- Led migration of monolithic video processing services to **microservices architecture**, improving deployment frequency and system reliability.
- Conducted architecture reviews for new video features, ensuring alignment with **large-scale systems** patterns and cost targets.
- Established **API design** standards for video metadata services, enabling cross-team integrations across the platform.
- Mentored 3 engineers on **distributed systems** design principles and **Go** performance optimization techniques.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built payment processing pipelines handling thousands of daily transactions with 99.9% success rate and PCI compliance.
- Architected real-time marketplace platform supporting thousands of concurrent live stream sessions with sub-second inventory updates.
- Led on-call rotation for production systems, triaging and resolving incidents affecting the live shopping experience.
- Designed **backend services** for bidder registration and authentication, reducing account creation time by 40%.
- Implemented **caching systems** for high-demand collectibles, achieving 95% cache hit rates during peak traffic.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Designed distributed caching layer for search results, reducing average query latency by 25% and cutting backend resource costs by millions annually.
- Led **Python** and **C++** codebase modernization initiative, reducing build times by 35% and improving type safety across 2M+ lines of code.
- Set technical direction for search backend improvements, coordinating with 3 cross-functional teams to ship features used by billions of users.
- Managed team of 6 engineers developing **search infrastructure** serving billions of queries across Google core search product.
- Architected key-value storage system managing **petabyte-scale** data across thousands of machines with 99.99% availability.

Software Engineer | Google | 02/2011 - 02/2015

- Developed ranking algorithms for search results, implementing features that improved click-through rates by 8% in offline evaluations.
- Built **distributed systems** for indexing and serving web documents at internet scale across multiple data centers.
- Optimized **database systems** queries handling billions of records, reducing p99 latency from 500ms to 80ms.
- Created monitoring dashboards for search infrastructure using internal observability tooling, enabling faster incident detection.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Implemented **backend services** APIs in **C++** and **Java** for message routing infrastructure processing millions of events daily.
- Developed backend services for enterprise communication products serving Fortune 500 customers at global scale.
- Participated in on-call rotation for production systems, resolving critical outages affecting enterprise customers.
- Designed **SQL** database schemas and query optimizations for high-throughput transaction processing systems.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Contributed to **C++** codebase refactoring effort that reduced memory usage by 15% in production systems.
- Developed testing frameworks for internal development tools, automating regression testing for core product components.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007